UMBC

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DATA 622 | Text Clustering and Similarity

1 message

Tony Diana <tonydiana1@gmail.com> Sun, Mar 29, 2026 at 7:55 PM
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Cc: Lisa Williams <ld530.williams@gmail.com>, "sravand1@umbc.edu" <sravand1@umbc.edu>

Hi everyone,

I will not be able to meet online on April 2. However, I have prepared material for you to review and complete at home. April 2 is your turn to stay home.

I have attached my presentation on text similarity and text clustering. Please review the lecture for further details.

T_DIANA Text Clustering & Similarity.mp4

I also have three exercises. I will send you the correction later.
1. Based on documents = ["The plaintiff filed a lawsuit for breach of contract.", "Contract disputes are common in civil litigation.", "The defendant denies all allegations of negligence.", "Personal injury and negligence claims require proof of harm."], create a Python program to assign the documents to two clusters.

2. How close is document 1 to 2 and document 1 to 3 based on the following contents:
doc1 = "The tenant failed to pay rent on time, violating the lease agreement."
doc2 = "The lease was breached because the monthly rent payment was late."
doc3 = "The company merged with a competitor to dominate the market."
3. Create an LLM code to measure the text similarity between doc1 and doc2, as well as doc1 and doc3.
doc1 = "The tenant failed to pay rent on time, violating the lease agreement."
doc2 = "The lease was breached because the monthly rent payment was late and not sent to the appropriate email."
doc3 = "The company merged with a competitor to dominate the market."

Use SentenceTransformer, not OpenAI or LangChain.

Please send the homework to Sravani. Thanks.

Dr. Tony Diana